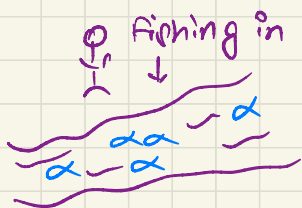
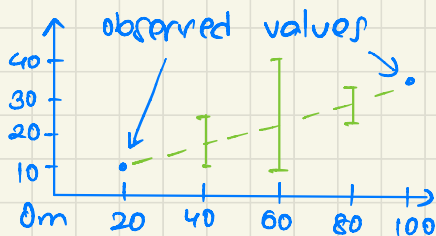


GAUSSIAN PROCESS REGRESSION



some areas have more fish than others



Goal: maximize # fish that you catch on that day

Problem: where along the river should you try?

Linear regression is fine but doesn't give us the uncertainty $\therefore$ GPR

- areas close to observed value (eg: 40m and 80m) have smaller uncertainty $\therefore$ know nearby point
- areas away from known data (eg: 60m) have larger uncertainty

Given known data (how far along in the river) x and $f(x)$ (# fish we catch)

$\hookrightarrow$

Find distribution of f for unknown points y

Assumption: $f(x), f(y) \sim \text{Normal}(\mu, \Sigma)$

fish for known points

fish for unknown points

How do we fill the covariance matrix?
kernel!

RBF kernel:

$$\Sigma_{ij} = \sigma^2 e^{-\frac{\|x_i - x_j\|^2}{2l^2}}$$

$$= \frac{\sigma^2 e^{-\frac{\|x_i - x_j\|^2}{2l^2}}}{2l^2}$$

As the dist. $\uparrow$, the covariance should $\downarrow$ $\therefore$ their $f(x)$ don't affect each other

other kernels:

- periodic
- linear

too much compared to x
where $f(x)$ are nearby

l : controls rate of decay
 σ : controls overall magnitude of covariance } hyperparameters

eg: `gpr_final = GaussianProcessRegressor(..., ..., ...,
kernel = 3.5**2 * RBF(length_scale = .355),
..., ..., ...)`
 $\downarrow$
 σ

if σ is bigger, we're expecting more uncertainty on average

$\Sigma f(20), f(30) = \Sigma f(90), f(100)$
 $\Sigma f(1), f(2)$ are same

Note: choice of kernel and σ and l do matter!

Pros:

- Uncertainty quantification
- Can capture complex relationships b/w inputs and outputs
- Interpretable $\therefore$ we can understand the model's behavior (unlike black box models)

Cons:

- Lose efficiency in high dimensional spaces
- Inverting the covariance matrix can be computationally expensive for large datasets
- Selecting appropriate kernel is crucial